Html colors

Html provides color to the text, to the background and to the webpage.

syntax

```
<font color = "colorname">
</font>
```

hexadecimal values for color

Red - #ff0000
green - #008000
Blue - #0000ff
Yellow - #ffff00
Black - #000000
white - #ffffff

background

```
<body bgcolor = "colrname">
```

CSS (cas style sheet)

It is a language use to control the appearance and layout of the web page

Why css is used

→ Change the text color, size and font
→ Change the background color or we can set the background image.
→ We can add the margin.
→ Create the responsive design that works on different screen sizes
→ add animations and visule efforts.

It as three types

1) Inline CSS
2) Internal CSS
3) External css

Inline css

In line css is written inside the style attribute of html element.

syntax:-

```
<tag style = " property : value;
               property : value;
               property : value; " > content </tag>
```

ex:-

```
<P style = " color : Red;
             font-family : Vardana.
             background-color : Black; " > Hello </P>
```

2) Internal css

Internal css is written inside the style tag within the head section of html document.

syntax:-

```
<head>
<style>
selector { property : value;
           property : value;
           property : value ; }
</style>
</head>
```

Ex: <head>
<style>
P { color: Red;
background-color: black;
font family: vardana;
}
</style>
</head>
<body> <P> Hellow </P> </body>

External CSS

External CSS is written in saperate .css file and link to the html file using link tag.

Syntax:
<head>
<link rel="stylesheet" href="file name">
</head>

Exbernd file. syntax
selector { property: value;
property: value;
property: value;
}

Ex:- <head>
<link rel="style sheet" href="r.css">
</head>
<body>
<p> hello </P>
</body>

r.CSS
P { color: Red;
font-family: vardena;
background-colov: Black;
}

lab program for responsive design
<!doctype html> ①
<html>
<head>
<title> Responsive design </title>
<meta name="veiwport" content="width=device-width,
initial-scale=1.0">
<style>
body { font-family: Arial;
text-align: center;
}
.container { display: flex;
gap: 20px;
}
.box { background-color: lightblue;
padding: 30px;
flex: 1;
}
@media (max-width: 600px).
.container {
flex-direction: column;
}
}
</style>
</head>
<body>
<h1> Responsive web page</h1>

Top right bottom left

```
<div class = "container">
  <div class = "box"> Box1 </div>
  <div class = "box"> box 2 </div>
  <div class = "Box"> box3 </div>
</div>
</body>
</html>
```

Responsive design

It is a web design technique which web page automatically adjust to the screen size of the device.

Class :- It is an html attribute used to give a name to an element or group of elements so that CSS can style them

ID

It is a html attribute. element use: to give a unique name to one specific element
→ uses dot in CSS
→ It uses # tag in ideas
→ It usally used only one unique element
ex:

```
<h1 id = "heading"> Responsive design </h1>
#heading { background-color : "Red";
           font-size : 20px;
         }
```

Write code create a simple table.

```
<!doctype html>
<html>
<head>
<title> Time table </title>
</hed>
<body>
<table border = "2" width = "700". align = "center">
<tr>
<th> days </th>
<th> 10 to 11 </th>
<th> 11 to 12 </th>
<th> 12 to 01 </th>
<th> 01 to 02 </th>
<th> 02 to 03 </th>
<th> 03 to 04 </th>
</tr> <tr>
<th> Monday </th>
<td> English </td>
<td> IA </td>
<td> Java </td>
<td rowspan = "3"> break time </td>
<td> IP </td>
<td> student activity </td>
</tr>
<tr>
<th> Tuseday </th>
<td> Java </td>
<td> AI </td>
<td> kannada </td>
<td> IP </td>
```

```
<td> IP lab </td>
</tr>
<tr>
<th> wendenday </th>
<td> lab </td>
<td> AI </td>
<td> kannada </td>
<td> PD </td>
<td> IP </td>
</tr>
</table>
</body>
</html>
```

Selectors.

selector	symbol	selects
Universal.	*	Everything
element/name	tagname (P)	All tag content.
Class	.box	element with class box
ID	#heading	Contents element with void #heading
group	P, h1, h2	It select multiple selector.

Ex:-
```
<!doctype html>
<html>
<head>
<title> selectors in CSS </title>
<style>
  * {  font-family: Arial;
  }
  P {  color: blue;
  }
  .box {  background-color: Yellow;
  }
  #heading {  color: red;
  }
  h1, P {  text-align: center;
  }
</style>
</head>
<body>
<h1 id="heading"> Welcome </h1>
<P> This is a paragraph </P>
<div class="box"> This a box </div>
</body>
</html>
```

CSS properties are the rules used to control appearance and layout of the html/ element on the web page

→ Each property has name and value.

Ex:- 1) Color, background-color, font-size, font-family width, height, border, margin, padding.

CSS colors

They can be specified in several ways

method	Example	result
color name	color: red.;	red text
HEX (Hexadecimal)	color: #FF0000;	red text
RGB (Red green blue)	color: rgb (225,0,0)	red text
Each values		

0-red
60-yellow
120-green
240-blue

Saturation → intensity.
Hue - basic color

→ RBnBA (Red green blue alpha) alpha (transference) (range b/w 0 to 1)	color: rgba(225.0,0,0.5);	50% transferent red text
→ HSL (Hue sachuration & Lightness) (Hue can be give value b/w 0 to 360) (S can be give value b/w 0 to 100%) (Lightness can be give value b/w 0 to 100%)	color: hsl(0, 50%, 40%); color: hsl(60, 50%, 70%);	red with 50% sachuration & 40% Lightness
→ HSLA (Alapha ideal with the transference) (valied in 0 to 1)	color: hsla(0, 50%, 40%, 0.5);	red with 50% sachuration & 40% Lightness & 50% transference

Css backgrounds
It deals with the background related properties,

property	discription	Example
background-color	set the background color	background-color: red;
background-image	set the background image	background-image: URL(" ");
background-repeat	controls the image repitation	background-repeat: repeat, /no-repeat;
background-position	sets the starting position of the image	background-position: center;
background-size	set the size of the image	background-size: cover/ contain / auto;

Short hand properties of Css background.
The background property allows you to write multiple background property in one line.
syntax:
selector {
background: color position /size repeate;
}

Css borders
It allows has to add border around html element
Ex:- selector {
border-style: solid;
border-width: 20px;
border-color: blue;
}
style :- solid, dotted, double, dash, groove

Shorthand property of border
syntax:
selector {
border: width style color;
}
Ex:- P { border: 5px dotted red; }

CSS margin
It is used to create space outside an elements border.
syntax:- selector { margin: value;
} Same margin on all side

selector { margin: 20px 40px; } // 1st value represent top and bottom
2nd value represent other two position

selector { margin: 20px 40px 10px; }
top, side (right), bottom

selector { margin: 20px 40px 10px 50px; }
top, Right, bottom, left

selector {
margin-top : 20px;
margin-right : 30px;
margin-bottom : 40px;
margin-left : 50px;
}

Padding in CSS

CSS Padding is space inside an element blw its content and its border.

Ex:- selector {
Padding-top : 20px
padding-right: 30px
padding-bttom: 40px
paddign-left : 50px
}
selector { padding : 20px 30px 40px 50px; }

CSS height and width.

It is used to control size of an element.

syntax:- selector { width: value;
height: value;
}

CSS font

font property is used to set the font style. for a text.

Ex:- selector { font-style:
font-size: 20px
font-family:
font-weight: 10
}

font-family :- It specifie type & face of a text
Ex: Arial, Helvetica, verdana, Georgia, fantasy

font-size : used to specify size of the text.

font-style :- It specifie style of the text.
value : normal, italic, oblique., bold

font-weight :- It specifie the thickness or boldness of the tex.
value: 100 → thin 400 → normal 500 - midium
600 - semibold 700 - bold 900 - extra bold

Short hand property of font.
selector {
font : font-style, font-weight, font-size, font-family; }

CSS text property.

It's provides properties to control the appearnce and alginment of text.

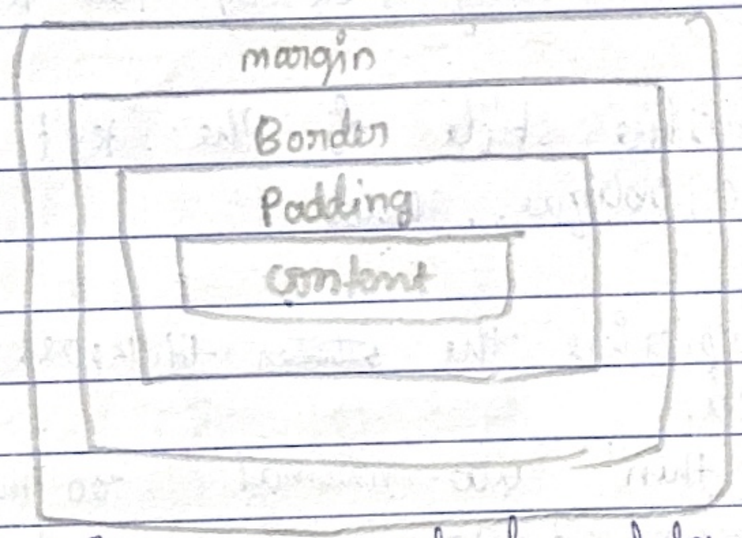
1) color
2) text-algin : centr, left, right, top, bottom
3) text-transform : change the letter case
uppercase / lowercase / capitalise

word-spacing: 10px 2px
letter-spacing:

CSS box modle

It discribes how every html element is represented as ~~a~~ rectanglues box on a web page. it consist of 4 main parts

1) containte content
2) Padding
3) Border
4) Marrgin



Content:- It is the actuale data text or image
Padding:- Space around the content or space b/w the content and border
Border :- Line surounding the padding and the content
margin :- space outside the border used to seperate the element.

Ex:-
```
<!doctype html>
<html>
<head>
<style>
P{ color: red;
   padding: 20px;
   border: 2px solid blue;
   margin: 50px;
}
P{ padding: 20px;
   border: 2px dotted black;
   margin: 30px
}
</style>
</head>
<body>
<P> Hi gm </P>
<h1> IP Class </h1>
</body>
</html>
```

Ex:-
```
<!doctype html>
<html>
<head>
<style>
P{ color: red;
   border: 20px dashed black;
   padding: 20px;
   margin: 50px;
   height: 100px;
   width: 200px;
}
h1{ color: pink
   border: 20px dotted black;
   padding: 20PX;
   margin: 30px
   height: 100px;
   width: 200px;
}
</style>
</head>
<body>
<P> Tom not to cry </P>
<h1> Be true to yourself </h1>
</body>
```

CSS flex box

It is the layout system that makes it easy to aline distrubute and size the elements with a container

→ By default ideams are arraged originantly left right.

→ It is used to arrage the element in row or column and control their allignmata and spacing.

step 1:- first use display : flex;

step 2:- flex direction as 4 values column(top to bottom row(left to right), row-reverse (right to left), column-reverse (bottom to top)

step 3:- Justify the content :. It controls the position of the items along with the main access.

Links in Html.

Links are created using anchor tag: <a>

Syntax

 link text

Link text means clickable text

Ex:-

 Visit google

 about

Frame in Html

Frames are used to divide the browser window into multiple sections with each section displaying different html document.

Ex- <!doctype html>
<html>
<head>
<title> Frame example </title>
</head>
~~<body>~~ <frameset cols="20%, 50%, 30%">
<frame src="menu.html">
<frame src="content.html">
<frame src="news.html">
</frameset>
</html>

menu.html

<!doctype html>
<html>
<head>
<title> menu </title>
</head>
<body>
<h2> Menu </h2>
 home
 about
 contact
</body>
</html>

content.html

<!doctype html>
<html>
<head>
<title> content </title>
</head>
<body>
 welcome to web site
</body>
</html>

news.html

<!doctype html>
<html>
<head>
<title> news </title>
</head>
<body>
<h2> Latest news </h2>
 news1
 news2
 news3
</body>
</html>

Frameset :- It is a tag used to divide the browser window into multiple parts

iframe

iframe is a html element used to embed another web page within current web page

→ iframe used to display another html page, website, video inside current web page

syntax

```
<!frame src = "Page.html" width = "500" height = "300">
</!frames>
```

Ex:-

```
<!doctype html>
<html>
<head>
<title> iframe </title>
</head>
<body>
<h1> Main web page </h1>
<p> This is main web page </p>
<iframe src = "welcome.html" height "500" width = "300"> </iframe>
</body>
</html>
```

```
<!doctype html>
<html>
<body>
<h2> welcome </h2>
<p> This is embeded page iframe </p>
</body>
</html>
```

Divide a webpage into two sections and insert iframe tage for each section and a load the external side as given below.

a) iframe1 . Youtube

b) iframe2 Google.

→

Semantic tag in html

Semantic tag in html are the tags that clearly describes the meaning and purpose of the content to both browser & developer

Tag	purpose
<Header>	defines the introduction of the Header area
<main>	defines the main content of the page
<figure>	contains an image / diagram
<figcaption>	Provides caption for the figure
<mark>	highlight the important text
<Nav>	define navigation links.
<section>	groups related content
<article>	defines independent content
<aside>	defines side content or related content.
<details>	provides additional information
<Summary>	defines visible heading for details
<footer>	defines footer area.

Multi media in html.

It mean using different types of media such as images audio and video.

Ex:-

```
<!doctype html>
<html>
<head>
<title> multimedia page</title>
<style>
img { width: 300px;
      border-radius: 10px;
    }
video { width: 400px
      }
</style>
</head>
<body>
< h2 > first multimedia page </h2>
<img src = "flower.jpg alt = "flower" >
<br><br>
<audio controls>
< source src = "song.mp3" type = "audio/mpeg" >
</audio>
<video>
< source src = "video.mp4" type = "video/mpy4" >
</video>
</body>
</html>
```

Html 5

It is a letest version of html used to create structured webpage and web applications.

features of Html 5

→ Semantic elements

→ multimedia support.

→ Convas :- used to draw graphics, animation & shapes

→ Geolocation :- allows webpage to determind the users geographical location with permission.

→ Reponsitive design supports.

→ improved forms.

→ It is introduce by 2014. → older version is introduce by 1999

html older version	html 5
→ multimedia requires external plagings / software	→ supports audio video directly.
→ fewer semantic tags	→ More semantic tags like header, foots, Nav
→ No canvas	→ Supports canvas.
→ limited form input types	→ provides new types like emile, date, number.
→ declaration type <html>	→ It is declaration type <!doctype html>